General Physics I Final Project Report

The Effects of Air Resistance and the Magnus Effect on the Projectile Motion of a Ping-Pong Ball

Topic: Comparison Between the Actual Flight Trajectory of a Ping-Pong Ball and the Ideal Projectile Model
Analysis Methods: Theoretical Derivation, Python Numerical Analysis, Graphical Comparison
Ball Specifications: Ping-Pong Ball (Diameter: 40 mm, Mass: 2.7 g)

Abstract

This study evaluates and compares the flight dynamics of a ping-pong ball across three distinct frameworks: the vacuum projectile model, the air resistance model, and the Magnus effect model. While standard physics curricula predominantly explore ideal projectile motion by neglecting atmospheric drag, real sports balls move through the air and are inherently subject to aerodynamic drag and rotational forces. Given its small diameter (40 mm) and exceptionally light mass (2.7 g), a ping-pong ball experiences highly pronounced air resistance effects.
In this project, governing equations of motion for each model were established based on Newton's laws of motion and numerically integrated using the Euler method in a Python environment. Simulation results revealed that the vacuum model produced the longest horizontal range and an ideal symmetric parabolic trajectory, whereas incorporating air resistance caused a drastic reduction in both horizontal range and peak height. Furthermore, when spin was incorporated into the Magnus model, the trajectory deviated from the standard two-dimensional x-y plane, exhibiting substantial lateral displacement along the z-axis. These findings demonstrate that accurately characterizing the real-world kinematics of a ball requires simultaneous consideration of gravity, aerodynamic drag, and rotational effects.

1. Introduction

1.1 Research Background

Projectile motion represents one of the most foundational concepts addressed in introductory general physics. Standard textbooks typically simplify this phenomenon by assuming a vacuum where only gravity acts on the object, thereby rendering its path a perfect parabola. Under this assumption, horizontal motion proceeds at a constant velocity while vertical motion undergoes uniform acceleration, allowing positions and velocities to be computed through straightforward kinematic equations.
In reality, objects travel through an atmospheric medium where air resistance acts opposite to the velocity vector, continuously draining kinetic energy. This drag force scales with velocity and is especially dominant for objects characterized by low mass and minimal inertia relative to their cross-sectional area. A ping-pong ball serves as an excellent candidate for observing these dynamics due to its physical properties—a diameter of 40 mm paired with a mass of just 2.7 g. Consequently, even when launched at identical initial velocities, a stark discrepancy emerges between the ideal vacuum model and real-world atmospheric flight. Furthermore, imparting spin onto a ping-pong ball induces a Magnus force via fluid-surface interactions, causing the ball to lift, drop rapidly, or curve laterally.

1.2 Research Objectives and Scope

The primary objective of this study is to critically evaluate whether a simple parabolic framework can accurately account for a ping-pong ball's motion and to quantify how air resistance and spin alter its true trajectory. To achieve this, a vacuum model, a drag model, and a Magnus model were implemented and simulated under equivalent or comparable baseline parameters.
The scope of this report is constrained strictly to the flight phase prior to ground impact. While a bounce model is integrated within the source code, it is treated as an optional structural extension since the core focus remains on the airborne trajectory. Additionally, due to a lack of empirical video tracking data, this report centers primarily on theoretical simulation design and numerical analysis.

2. Theoretical Background

2.1 Parabolic Motion in a Vacuum

In a vacuum devoid of air resistance, gravity is the sole force acting on a projectile. Defining the horizontal axis as x and the vertical axis as y, the equations of motion are expressed as follows:

```
x(t) = x0 + v0x * t,      y(t) = y0 + v0y * t - (1/2) * g * t²
```

Setting the initial position at the origin yields $x(t) = v0 * \cos(\theta) * t$ and $y(t) = v0 * \sin(\theta) * t - (1/2) * g * t^2$. Eliminating the time variable t transforms y into a quadratic function of x, defining a parabolic path. Because horizontal velocity remains unchanged, this model predicts the absolute maximum possible range for any given initial velocity.

2.2 Air Resistance

Objects moving through fluids encounter an aerodynamic drag force. For a spherical object, this drag force is generally formulated as:

```
Fd = -(1/2) * Cd * ρ * A * |v| * v
```

Where Cd represents the drag coefficient, ρ is the air density, A is the cross-sectional area of the ball, and v is the velocity vector. The negative sign signifies that drag acts directly counter to the direction of motion. In the simulation code, Cd is defined as a velocity-dependent function, meaning that acceleration is non-uniform and varies continuously. Drag progressively degrades horizontal velocity and restricts both ascending and descending vertical speeds, shortening peak altitude, flight duration, and overall range.

2.3 The Magnus Effect

A rotating sphere experiences a lift force known as the Magnus force, which acts perpendicular to both the spin axis and the velocity vector. It can be simplified as:

```
Fm = CL * π * R³ * ρ * (ω × v)
```

Where CL is the lift coefficient, R is the radius of the ball, and ω is the angular velocity vector. Depending on the vector cross product $\omega \times v$, the ball will curve upward, downward, or sideways. This physical effect explains how topspin, backspin, and sidespin drastically alter trajectories in table tennis.

2.4 Numerical Methods

When air resistance and Magnus forces are factored in, the net force depends dynamically on velocity, rendering analytical solutions intractable. Thus, this study employs the Euler method, which iteratively updates acceleration, velocity, and position over small, discrete time steps (*dt*):

```
v(t + dt) = v(t) + a(t) * dt
r(t + dt) = r(t) + v(t + dt) * dt
```

A time step of *dt = 0.001 s* was used in this simulation. This interval is sufficiently small to approximate the true differential equations with high accuracy.

3. Methodology

3.1 Simulation Conditions

Simulations were conducted using Python's NumPy and Matplotlib libraries. The ball parameters were modified from the default script to match standard table tennis specifications: a diameter of 40 mm, radius of 0.020 m, and mass of 0.0027 kg. Air density was set to 1.2 kg/m³, and gravitational acceleration was defined as 9.81 m/s².

Parameter	**Value**
Ball Type	Ping-Pong Ball
Diameter	40 mm
Radius (R)	0.020 m
Mass (m)	0.0027 kg
Air Density (ρ)	1.2 kg/m³
Gravitational Acceleration (g)	9.81 m/s²
Time Step (dt)	0.001 s

Table 1. Baseline Physical Quantities Used in Simulation

3.2 Model Configurations

Three models were constructed for comparative analysis: the Vacuum model (gravity only), the Air Drag model (gravity + drag), and the Magnus Spin model (gravity + drag + rotational forces). To isolate the direct impact of drag, the Vacuum and Air Drag models utilized identical initial velocities of (20, 10, 0) m/s. For the Magnus model, initial conditions were set to an initial velocity of (20, 5, 2) m/s and an angular velocity of (0, 46, 0) rad/s to explicitly observe lateral deviations.

Model	Forces Included	Initial Conditions
Vacuum	Gravity	v0 = (20, 10, 0) m/s
Air Drag	Gravity + Air Resistance	v0 = (20, 10, 0) m/s
Magnus Spin	Gravity + Drag + Magnus Force	v0 = (20, 5, 2) m/s, ω = (0, 46, 0) rad/s

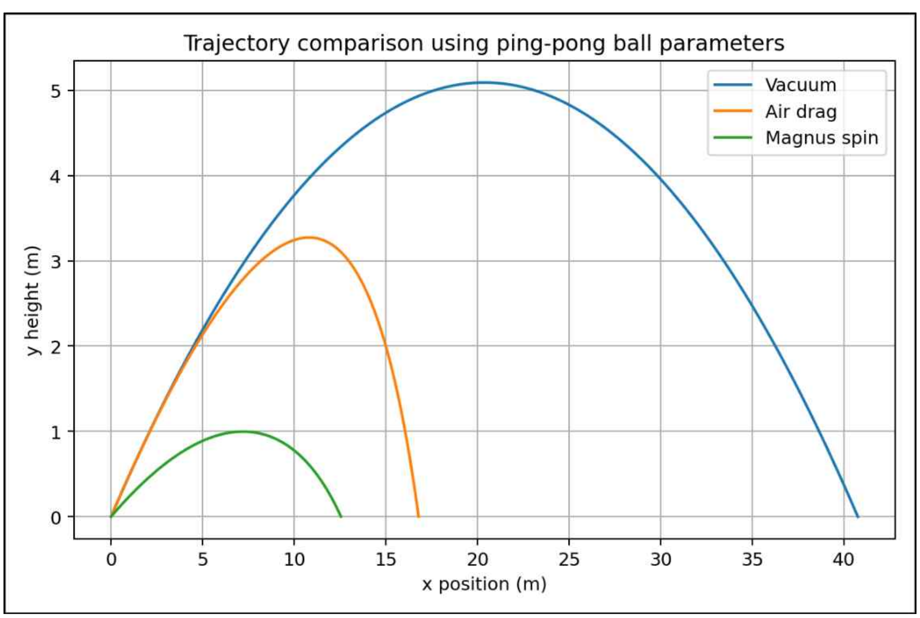
Table 2. Overview of the Three Kinematic Models

3.3 Code Adaptation and Implementation

The program calculates the net forces based on the instantaneous velocity at each step, derives the resulting acceleration by dividing the net force by the ball's mass, and updates the velocity and position arrays sequentially. The execution terminates immediately when the vertical position y drops below 0, indicating ground impact. Outputs are visualized via three plots: an x-y trajectory comparison, an x-z lateral displacement profile, and a time-height curve. The baseline template code originally defaulted to dimensions resembling a soccer ball. Consequently, all tracking variables were updated to R = 0.020 m and m = 0.0027 kg to properly reflect a ping-pong ball. It is worth noting that the drag coefficient routine, labeled cd_soccer, is empirical soccer ball data. While it may not perfectly represent the precise fluid profile of a table tennis ball, the simulation effectively serves as a structural model experiment to illustrate the relative qualitative trends of drag and spin on light spheres.

4. Results

4.1 Trajectory Analysis



As illustrated in the trajectory data, the Vacuum model displays a perfectly symmetrical parabola and spans the maximum horizontal distance. Conversely, the Air Drag model features a visibly stunted trajectory due to ongoing energy dissipation during both the ascent and descent phases, significantly lowering both maximum altitude and final range. Because the Magnus model was initialized with a different vertical velocity to accommodate cross-axis spin, its curve is interpreted primarily to evaluate structural geometric changes introduced by the side-acting lift vector rather than direct peak-to-peak height comparisons.

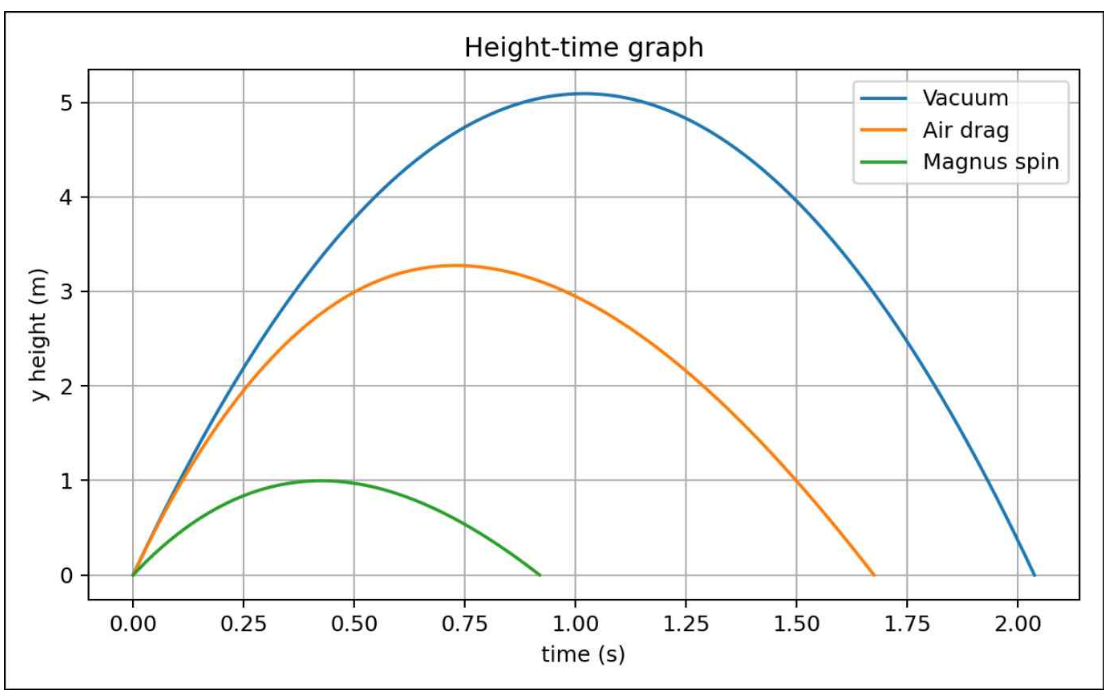
4.2 Quantitative Comparison

Model	Flight Time (s)	Horizontal Range (m)	Max Height (m)	Final z-Displacement (m)
Vacuum Model	2.038	40.760	5.092	0.000
Air Drag Model	1.675	16.784	3.273	0.000
Magnus Model	0.919	12.546	0.997	-1.041

Table 3. Quantitative Simulation Output Summary

Table 3 consolidates the quantitative flight data. The Vacuum and Air Drag models can be compared directly since they share identical launch vectors. Introducing air drag cuts the horizontal range by more than half (dropping from 40.760 m to 16.784 m) and reduces peak altitude from 5.092 m to 3.273 m. This drastic change underscores how severely a low-mass object like a table tennis ball is decelerated by fluid resistance.

4.3 Height-Time Dynamics



The temporal height profile demonstrates that the Vacuum model maintains a perfectly symmetric curve due to its constant gravitational acceleration. In contrast, the Air Drag model displays an asymmetric curve with a shallower apex and a compressed flight duration, as drag constantly opposes velocity and accelerates the decay of vertical momentum during ascent while delaying descent speeds.

4.4 Lateral Displacement

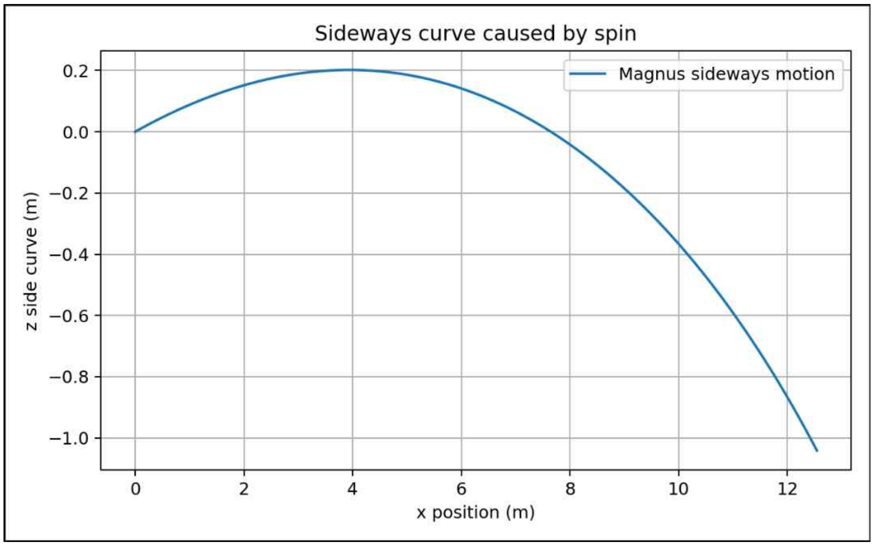


Figure 3 isolates the z-axis position relative to the x-axis forward progress within the Magnus framework. With the spin vector aligned along the y-axis, the cross product $\omega \times v$ drives a consistent lateral force. The resulting curve confirms that a spinning projectile breaks its 2D planar constraints, inducing a highly visible sideways drift that reaches a final displacement of -1.041 m.

5. Discussion

5.1 Impact of Atmospheric Drag

The vacuum model severely overestimates flight range because it assumes zero horizontal deceleration. In a true atmospheric setting, drag acts continuously against the flight vector. Because a ping-pong ball possesses very little mass, a moderate drag force produces a remarkably high negative acceleration ($a = F/m$). Consequently, ignoring air resistance yields completely unphysical trajectories for lightweight sports balls.
Drag also distorts the vertical flight path. On the way up, drag points downwards, reinforcing gravity and causing the ball to lose upward velocity more rapidly. On the descent, drag points upwards, actively opposing gravity and slowing the fall. This demonstrates how fluid resistance continuously works to dissipate mechanical energy throughout the entire timeline of motion.

5.2 Implications of the Magnus Effect

The Magnus effect is indispensable for modeling realistic ball behavior in modern table tennis. Heavy topspin causes a ball to dive sharply, while backspin alters the descent profile, entirely due to the shifting direction of the lift vector. This simulation successfully replicates this behavior; as soon as spin is enabled, a notable z-axis displacement emerges, creating a dynamic three-dimensional trajectory.
However, this model relies on a highly idealized Magnus formulation. Real-world Magnus forces depend tightly on surface roughness, exact spin rates, localized airflow separation, Reynolds numbers, and boundary layer transitions. Acquiring empirical spin rates and positional coordinate streams would allow for a much more precise calibration of these coefficients.

5.3 Limitations and Error Analysis

First, the drag coefficient routine is not natively optimized for a ping-pong ball. The cd_soccer function in the source code relies on aerodynamic profiles matching a soccer ball, which differs in surface texture and scale. Second, the environmental simulation assumes a completely static atmosphere with uniform air density, neglecting ambient indoor drafts and tracking parallax errors common in real setups.
Third, while the Euler method is straightforward to execute, it can accumulate numerical drift if the time step is too large. Although $dt = 0.001\ s$ was small enough to mitigate massive errors here, implementing higher-order schemes like the Runge-Kutta 4th order (RK4) method would yield better long-term stability. Lastly, the chosen lift coefficient ($CL = 0.9$) and angular velocity (*46 rad/s*) are merely illustrative baselines and may deviate from specific athletic conditions.

5.4 Future Work and Extensions

Subsequent iterations of this project should utilize high-speed videography paired with marked ping-pong balls to capture authentic spin rates and empirical launch states. Employing video tracking suites like Tracker would allow researchers to extract coordinate data over time to cross-validate simulations directly against real physical paths. Replicating trials with non-spinning and spinning balls under identical launch parameters would successfully isolate drag profiles from Magnus forces.

6. Potential Extensions: The Bounce Model

The underlying source code includes a functional bounce calculation routine. This module utilizes pre-impact translational and rotational velocities to determine post-impact kinematic states. The vertical velocity is inverted via a coefficient of restitution (ey), while horizontal velocity and spin are cross-coupled by surface friction coefficients and the ball's moment of inertia. Although this report focuses entirely on the airborne phase, integrating this bounce routine would seamlessly extend the study to map complex post-impact paths across a table tennis surface.
In competitive play, ball spin and velocity undergo major shifts immediately upon table contact. Incorporating a robust bounce framework is the logical next step toward capturing real-world sports analytics. However, because impact dynamics involve highly complex variables—such as surface friction, elastic deformation timelines, and momentary contact dwell times—systematic empirical tracking is necessary to determine accurate physical constants.

7. Conclusion

This study systematically evaluated the kinematics of a ping-pong ball across vacuum, air drag, and Magnus spin models. The vacuum model, which solely accounts for gravitational acceleration, yields an idealized symmetric trajectory and an overestimated flight range. Introducing air drag causes a massive reduction in both horizontal range and peak altitude due to rapid velocity attenuation.
Furthermore, adding angular velocity activates the Magnus force, generating notable lateral deviations that shift the flight path into three dimensions. This directly illustrates the physical mechanics underlying the drastic trajectory changes observed in competitive sports. Ultimately, the real flight of a ball must be treated not as a simple parabola, but as a complex system governed by the simultaneous interplay of gravity, fluid drag, and rotational mechanics.
In summary, this project successfully bridges core textbook projectile mechanics with real-world sports applications. Combining theoretical equations with numerical Python simulations highlights the critical variations separating ideal physics models from real physical phenomena.

References

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